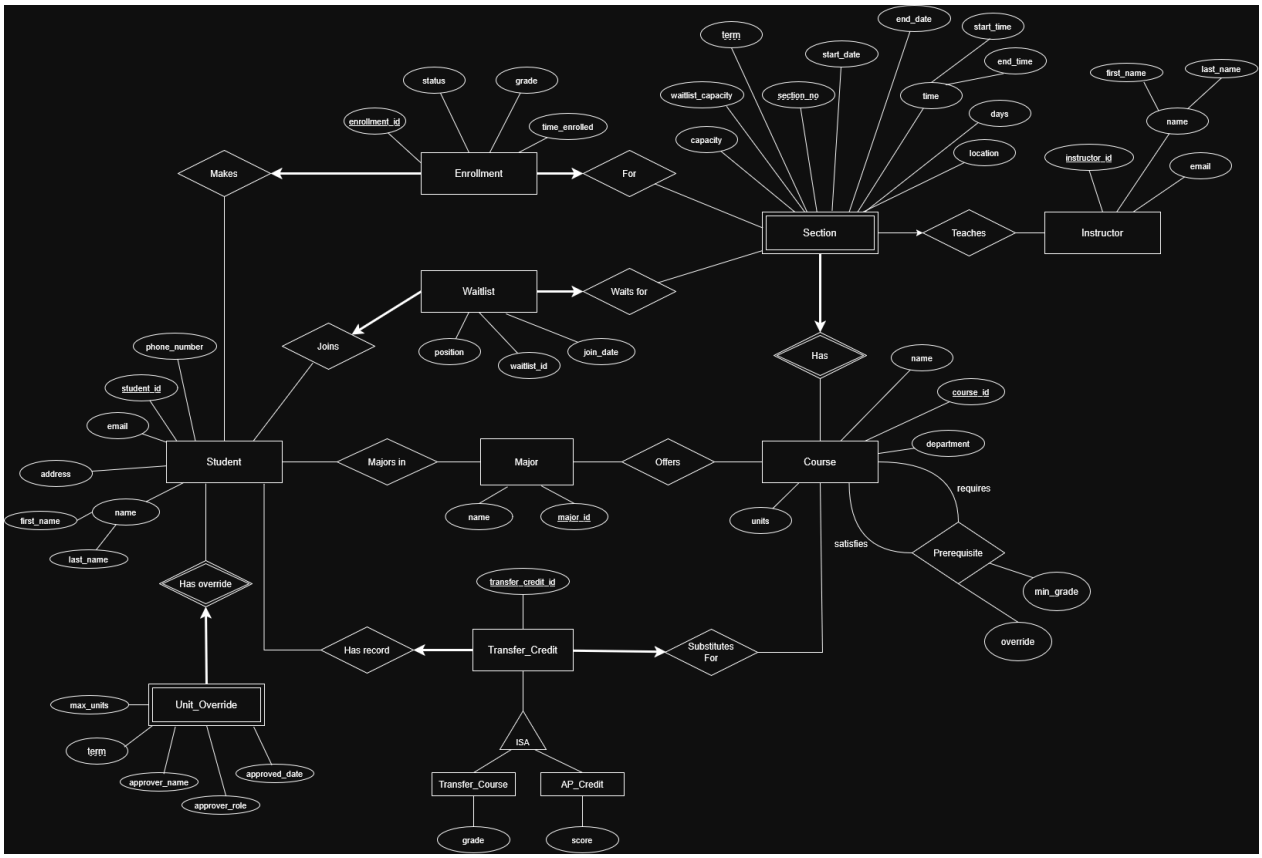
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Project Enrollment
8 November 2026

ER Diagram



Relational Mapping

Relation	ON DELETE
Student(**student_id**, first_name, last_name, address, phone_number, email)	
Enrollment(**enrollment_id**, student_id, course_id, term, section_no, status, grade, time_enrolled)	FOREIGN KEY (student_id) REFERENCES Student(student_id) ON DELETE CASCADE FOREIGN KEY (course_id, term, section_no) REFERENCES Section(course_id, term, section_no) ON DELETE CASCADE

Waitlist(**waitlist_id**, student_id, course_id, term, section_no, position, join_date)	FOREIGN KEY (student_id) REFERENCES Student (student_id) ON DELETE CASCADE FOREIGN KEY (course_id, term, section_no) REFERENCES Section(course_id, term,section_no) ON DELETE CASCADE
Course(**course_id**, name, units, department)	
Section(**course_id, term, section_no,** start_date, end_date, capacity, waitlist_capacity, days, start_time, end_time, location, instructor_id)	FOREIGN KEY (course_id) REFERENCES Course(course_id) ON DELETE CASCADE FOREIGN KEY (instructor_id) REFERENCES Instructor (instructor_id) ON DELETE SET NULL
Instructor(**instructor_id**, first_name, last_name, email)	
Unit_Override(**student_id**, **term**, max_units, approver_name, approver_role, approved_date)	FOREIGN KEY (student_id) REFERENCES Student(student_id) ON DELETE CASCADE
Major(**major_id**, name)	
Transfer_Course(**transfer_credit_id**, grade, student_id, course_id)	FOREIGN KEY (student_id) REFERENCES Student(student_id) ON DELETE CASCADE FOREIGN KEY (course_id) REFERENCES Course(course_id) ON DELETE SET NULL
AP_Credit(**transfer_credit_id**, score, student_id, course_id)	FOREIGN KEY (student_id) REFERENCES Student(student_id) ON DELETE CASCADE FOREIGN KEY (course_id) REFERENCES Course(course_id) ON DELETE SET NULL

Major_Course_Offering(major_id, course_id)	FOREIGN KEY (major_id) REFERENCES Major(major_id) ON DELETE CASCADE FOREIGN KEY (course_id) REFERENCES Course(course_id) ON DELETE CASCADE
Student_Major(student_id, major_id)	FOREIGN KEY (student_id) REFERENCES Student(student_id) ON DELETE CASCADE FOREIGN KEY (major_id) REFERENCES Major(major_id) ON DELETE RESTRICT
Course_Prerequisite(course_id, prereq_course_id, min_grade, override)	FOREIGN KEY (course_id) REFERENCES Course(course_id) ON DELETE CASCADE FOREIGN KEY (prereq_course_id) REFERENCES Course(course_id) ON DELETE RESTRICT

FD's
1. (student_id) → (first_name, last_name, address, phone, email)
2. (course_id) → (course_name, units, department)
3. (course_id, section_no, term_name) → (capacity, waitlist_capacity, start_date, end_date, days, start_time, end_time, location, instructor_id)
*section overlap taught by instructors should be a constraint.
4.
(student_id, section_id) → (status, grade, time_enrolled)
(enrollment id) → (student id, section id, status, grade, time enrolled)
5.
(instructor_id) → (first_name, last_name)
6.
(course_id, prerequisite_course_id) → (minimum_grade, override)
*ignored substitutes for now
7. (student_id, section_id, term_name) → (join_date, position)
8. (student_id, term_name) → (max_units, override)
9.
*Schedule constraint to handle overlap.
Additional Primary Keys:
1. (student id, major_id)
2. (course_id, major_id)